Dugoff Tire Model for Mining Haul Trucks

Mathematical Derivation and Implementation

Mario Tilocca

January 21, 2026

Abstract

This document presents the mathematical foundation of the Dugoff tire model for application to autonomous mining haul trucks operating in Western Australian desert conditions. The Dugoff model provides a computationally efficient representation of tire-ground interaction that captures friction saturation on loose surfaces while maintaining analytical tractability. We derive the complete model equations and provide parameter identification for the XCMG XDE320 electric dump truck (320-ton payload).

1 Introduction

The Dugoff tire model [**?**] is selected for mining haul truck simulation due to its key advantages:

- Handles combined longitudinal and lateral slip (critical for cornering while accelerating/braking)
- Models friction saturation on loose surfaces (gravel, iron ore dust)
- Closed-form equations (no iterative solvers required)
- Captures smooth transition from linear to sliding regimes

2 Fundamental Equations

2.1 Tire Force Model

The Dugoff model expresses tire forces as:

$$F_x = C_x \sigma_x f(\lambda) \tag{1}$$

$$F_y = C_y \sigma_y f(\lambda) \tag{2}$$

where:

- F_x = longitudinal tire force [N]
- F_y = lateral tire force [N]
- C_x = longitudinal slip stiffness [N]
- C_y = lateral slip stiffness [N]
- σ_x = longitudinal slip ratio [-]
- σ_y = lateral slip ratio [-]
- $f(\lambda)$ = friction saturation function
- λ = friction utilization parameter

2.2 Slip Ratio Definitions

Longitudinal Slip Ratio:

$$\sigma_x = \frac{\omega R - V_x}{V_x} \tag{3}$$

where ω is wheel angular velocity [rad/s], R is tire effective rolling radius [m], and V_x is vehicle longitudinal velocity [m/s].

Sign Convention:

- $\sigma_x > 0$: Driving (wheel faster than vehicle)
- $\sigma_x < 0$: Braking (wheel slower than vehicle)
- $\sigma_x = 0$: Free rolling

Lateral Slip Ratio:
For small slip angles (typical in normal driving):

$$\sigma_y = \frac{V_y}{V_x} \approx \tan(\alpha) \approx \alpha \tag{4}$$

where V_y is lateral velocity at tire contact patch [m/s] and α is tire slip angle [rad].

2.3 Friction Utilization Parameter

The friction utilization parameter λ determines whether the tire operates in the linear or friction-limited regime:

$$\lambda = \frac{\mu F_z}{2\sqrt{C_x^2 \sigma_x^2 + C_y^2 \sigma_y^2}} \tag{5}$$

where:

- μ = coefficient of friction (surface-dependent)
- F_z = normal load on tire [N]

Physical Interpretation:

- $\lambda \geq 1$: Linear regime (no sliding, forces proportional to slip)
- $\lambda < 1$: Nonlinear regime (friction-limited, approaching saturation)
- $\lambda \to 0$: Complete sliding (forces approach zero)

2.4 Friction Saturation Function

The saturation function $f(\lambda)$ is defined piecewise:

$$f(\lambda) = \begin{cases} (2-\lambda)\lambda = 2\lambda - \lambda^2, & \lambda < 1 \\ 1, & \lambda \geq 1 \end{cases} \tag{6}$$

This function ensures:

1. Continuity at $\lambda = 1$: $\lim_{\lambda \to 1^-} f(\lambda) = 1$
2. Smooth transition from linear to saturated regime
3. Maximum value of $f(\lambda) = 1$ at $\lambda = 1$
4. Monotonic decrease as $\lambda \to 0$

3 Force Saturation Proof

Theorem: The Dugoff model guarantees that total tire force never exceeds available friction:

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z \tag{7}$$

Proof:
Start with the total force magnitude:

$$\begin{aligned}\sqrt{F_x^2 + F_y^2} &= \sqrt{(C_x\sigma_x f(\lambda))^2 + (C_y\sigma_y f(\lambda))^2} &(8)\\ &= f(\lambda)\sqrt{C_x^2\sigma_x^2 + C_y^2\sigma_y^2} &(9)\end{aligned}$$

For $\lambda < 1$ (friction-limited regime), substitute $f(\lambda) = 2\lambda - \lambda^2$ and λ from Eq. (5):

$$\begin{aligned}\sqrt{F_x^2 + F_y^2} &= (2\lambda - \lambda^2)\sqrt{C_x^2\sigma_x^2 + C_y^2\sigma_y^2} &(10)\\ &= (2\lambda - \lambda^2)\cdot\frac{\mu F_z}{2\lambda} &(11)\\ &= (2-\lambda)\,\frac{\mu F_z}{2} &(12)\\ &< \mu F_z \quad (\text{since } 0 < \lambda < 1) &(13)\end{aligned}$$

For $\lambda \geq 1$ (linear regime), $f(\lambda) = 1$, so the tire is not friction-limited and the inequality holds by construction. □

4 Parameter Identification

4.1 Tire Slip Stiffness

For 40.00R57 mining tires on XCMG XDE320:

Base Values at Reference Load:

$$\begin{aligned}C_{x,\text{base}} &= 280{,}000\ \text{N} &(14)\\ C_{y,\text{base}} &= 220{,}000\ \text{N} &(15)\\ F_{z,\text{ref}} &= 800{,}000\ \text{N (800 kN)} &(16)\end{aligned}$$

Load-Dependent Scaling:
Slip stiffness increases with normal load following an empirical square-root relationship:

$$\begin{aligned}C_x(F_z) &= C_{x,\text{base}}\left(\frac{F_z}{F_{z,\text{ref}}}\right)^{0.5} &(17)\\ C_y(F_z) &= C_{y,\text{base}}\left(\frac{F_z}{F_{z,\text{ref}}}\right)^{0.5} &(18)\end{aligned}$$

This nonlinear scaling captures the stiffening effect of increased contact patch pressure.

4.2 Friction Coefficients

Surface-dependent friction coefficients for Pilbara mining conditions:

Note: Peak friction μ_{peak} applies at low slip ratios ($|\sigma_x| < 0.05$). Slide friction μ_{slide} applies when wheels are locked ($|\sigma_x| \approx 1$). The Dugoff model naturally interpolates between these values.

Surface Type	μ_{peak}	μ_{slide}
Dry pavement (smooth haul road)	0.85	0.75
Gravel compact (baseline)	0.72	0.65
Gravel loose (after grading)	0.55	0.48
Iron ore dust (dry)	0.45	0.38
Iron ore dust (wet)	0.30	0.25
Mud (wet season)	0.25	0.20
Sand (loose)	0.40	0.35

Table 1: Friction coefficients by surface type

4.3 Normal Load Distribution

For XCMG XDE320 (empty weight = 218 tons, GVW = 538 tons):

Static Load Distribution:

$$F_{z,\text{front}} = \frac{mg \cdot d_{\text{rear}}}{L} \tag{19}$$

$$F_{z,\text{rear}} = \frac{mg \cdot d_{\text{front}}}{L} \tag{20}$$

where:

- m = vehicle mass [kg]
- $g = 9.81$ m/s^2
- $L = 6.3$ m (wheelbase)
- d_{front} = distance from CoG to front axle [m]
- d_{rear} = distance from CoG to rear axle [m]

Dynamic Load Transfer (Longitudinal):

During acceleration or braking, load transfers between axles:

$$\Delta F_z = \frac{m a_x h_{\text{CoG}}}{L} \tag{21}$$

$$F_{z,\text{front}} = F_{z,\text{front,static}} - \Delta F_z \tag{22}$$

$$F_{z,\text{rear}} = F_{z,\text{rear,static}} + \Delta F_z \tag{23}$$

where a_x is longitudinal acceleration [m/s^2] and h_{CoG} is center of gravity height [m].

Sign Convention:

- $a_x > 0$: Acceleration (weight transfers to rear)
- $a_x < 0$: Braking (weight transfers to front)

5 Numerical Implementation

5.1 Algorithm

For each tire at each timestep:

1. Compute slip ratios σ_x, σ_y from kinematics
2. Compute normal load F_z from load distribution
3. Compute slip stiffness $C_x(F_z)$, $C_y(F_z)$ from load scaling
4. Compute friction utilization λ from Eq. (5)
5. Compute saturation function $f(\lambda)$ from Eq. (6)
6. Compute tire forces F_x, F_y from Eqs. (1)-(2)

5.2 Numerical Stability

Division by Zero Protection:

When computing slip ratios at low speeds:

$$\sigma_x = \frac{\omega R - V_x}{\max(V_x, V_{\min})} \tag{24}$$

where $V_{\min} = 0.5$ m/s prevents division by zero.

Epsilon in Lambda Calculation:

To avoid division by zero when slip is zero:

$$\lambda = \frac{\mu F_z}{2\sqrt{C_x^2\sigma_x^2 + C_y^2\sigma_y^2} + \epsilon} \tag{25}$$

where $\epsilon = 10^{-6}$ N.

6 Expected Behavior

6.1 Traction-Limited Acceleration

For XCMG XDE320 loaded (538 tons) on gravel compact ($\mu = 0.72$):

$$F_{z,\text{rear}} \approx 0.75 \times 538{,}000 \times 9.81 = 3{,}956{,}000 \text{ N} \tag{26}$$

$$F_{x,\max} = \mu F_{z,\text{rear}} = 0.72 \times 3{,}956{,}000 = 2{,}848{,}000 \text{ N} \tag{27}$$

$$F_{x,\text{motor}} = \frac{T_{\text{motor}}}{R} = \frac{145{,}000}{1.93} = 75{,}100 \text{ N} \tag{28}$$

Result: Motor force $\ll$ traction limit, so truck is **power-limited**, not traction-limited.

6.2 Emergency Braking

For same truck on wet iron ore dust ($\mu = 0.30$):

$$F_{x,\max} = 0.30 \times 538{,}000 \times 9.81 = 1{,}585{,}000 \text{ N} \tag{29}$$

$$F_{x,\text{brake}} = \frac{T_{\text{brake}}}{R} = \frac{180{,}000}{1.93} = 93{,}300 \text{ N} \tag{30}$$

Result: Still power-limited, but much closer to traction limit. On loose gravel ($\mu = 0.55$), wheels would lock.

7 Validation

7.1 Unit Tests

- Verify $f(\lambda)$ continuity at $\lambda = 1$
- Confirm $\sqrt{F_x^2 + F_y^2} \leq \mu F_z$ for random inputs
- Test load transfer during acceleration/braking

7.2 Integration Tests

- Straight-line acceleration on various surfaces
- Emergency braking with ABS simulation
- Cornering at limit friction
- Combined braking and cornering (friction circle)

8 Conclusion

The Dugoff tire model provides a computationally efficient framework for simulating mining haul truck dynamics on loose surfaces. Its closed-form equations and guaranteed friction saturation make it suitable for real-time simulation and control algorithm development.

Future work includes:

- Validation against field data from XCMG XDE320
- Extension to 3D terrain (grades, banking)
- Thermal effects on friction coefficients
- Comparison with Pacejka Magic Formula

References

[1] Dugoff, H., Fancher, P. S., and Segel, L., *An Analysis of Tire Traction Properties and Their Influence on Vehicle Dynamic Performance*, SAE Technical Paper 700377, 1970.

[2] Pacejka, H., *Tire and Vehicle Dynamics*, 3rd Edition, Butterworth-Heinemann, 2012.

[3] Rajamani, R., *Vehicle Dynamics and Control*, 2nd Edition, Springer, 2012.